

Product Requirements Document (PRD)

DenkKern — Lena 2.0

AI-Assisted Shipment Disruption Decision Support

1. Product Overview

Product Vision

DenkKern helps manufacturing and logistics teams make faster, financially informed decisions when shipments are disrupted. Instead of only predicting delays, the system evaluates operational scenarios, estimates financial impact, and recommends actions to minimize business loss.

Problem Statement

Manufacturing firms often rely on critical shipments that can be delayed due to maritime disruptions, port congestion, strikes, or weather events.

Current systems provide: tracking alerts ETA updates But they do not answer the key business question:

"What should we do now?"

As a result: decisions are reactive production downtime increases financial losses escalate

2. Use Case

Primary Persona: Lena

Production Manager at a Hamburg shipbuilding company.
She depends on marine-quality bolts arriving on time to maintain production.

Scenario

A shipment traveling from Asia to Hamburg faces potential disruption near France and possible delays in Amsterdam and Hamburg.

Every day of delay causes: €150,000 production loss Lena must decide whether to: wait expedite order replacement parts

3. Product Goal

The system should: Predict shipment delay risk Quantify uncertainty Generate decision scenarios Estimate business impact Recommend the financially optimal response

4. Core Value Proposition

DenkKern transforms: shipment uncertainty into actionable business decisions

5. Product Scope (MVP)

A. Shipment Monitoring vessel tracking ETA prediction disruption monitoring **B. Risk Prediction** expected arrival optimistic/pessimistic arrival delay probability confidence score **C. Decision Scenarios** Wait for shipment Expedite from intermediate port Order replacement parts **D. Financial Impact Estimation** action cost production loss total expected cost **E. Recommendation Engine** best option expected savings reasoning **F. Decision Dashboard** shipment status risk level scenario comparison recommendation

6. Out of Scope (MVP)

automatic execution ERP write-back dynamic procurement full workflow builder custom ontology platform autonomous AI agents

7. User Flow

System detects disruption signals

Examples: maritime incidents, congestion, strikes, weather

Prediction model estimates

ETA, delay probability, confidence interval

Decision engine generates scenarios

Cost engine estimates

production loss, mitigation cost, expected total cost

UI presents

ranked options, recommendation, financial impact

8. Functional Requirements

Shipment Intelligence System shall ingest vessel location data. System shall monitor disruption signals. System shall calculate expected arrival time. **Risk Modeling** System shall calculate optimistic ETA, pessimistic ETA, expected delay. System shall estimate delay probability. **Scenario Engine** System shall support multiple response options. Each option shall include delay estimate, action cost, production loss, total expected cost. **Recommendation Engine** System shall rank options by expected financial outcome. System shall generate plain-language explanations. **User Interface** Dashboard shall display shipment status, risk indicators, scenario cards, recommendation panel. User shall be able to select an action scenario.

9. Non-Functional Requirements

UI response < 2 seconds Prediction refresh every 15 minutes Graceful degradation if external APIs fail Recommendation reasoning must be visible Role-based access Encrypted API communication

10. Technical Architecture

External Data Sources AIS vessel data weather APIs port congestion feeds disruption alerts **ML Prediction Layer** ETA uncertainty delay probability **Decision Engine** operational scenarios financial comparison **Recommendation Layer** expected cost risk **Frontend Dashboard** scenarios recommendation risk explanation

11. Example Decision Output

Option	Delay	Total Cost	Risk
Wait	5 days	€750k	High
Expedite	2 days	€500k	Medium
Replacement Parts	0 days	€500k	Low

Recommended Action: Order replacement parts from Poland.

Reason: avoids production downtime minimizes expected loss reduces operational risk

12. Success Metrics

Operational reduction in downtime losses faster decision time **Product** scenario interaction rate recommendation acceptance rate **Business** pilot customer conversion paid PoC deployments

13. Future Roadmap

Phase 2 ERP integration procurement workflows multi-shipment optimization **Phase 3** broader supply chain orchestration CBAM cost intelligence strategic sourcing optimization

14. Key Product Principle

DenkKern is not a tracking dashboard.

It is a decision-support system for operational disruptions.